Daily Drilling Report
SPD19B-03
2/21/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
2.0	32"	617.00	72.83	31.70	96.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
26" CP, 178.00mKB					1.42
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
163.00	183.00		20.00	1.20	16.7

Operations at Report Time

Operations Summary

Cont' drill 32" hole f/ 163m to 183m (TD). Sweep hole and perform wiper trip. POOH BHA#1 to surface. Run and CMT 26" Conductor Pipe. Prepare for Skid f/ SPD19B-03 to SPD19B-05.

Operations next 24hrs

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Vahab Kari	Night Drilling Sup.
Behnam Jahangiri	Drilling Engineer
Mohammad Hasan Hemmati	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill		
BOP Test		
Chemical Spill Drill		
Fire Drill		
H2S Drill		
Heli Crash Drill	2/20/2014	5/2/2014
Medical Drill		
MOB Drill		
Pit Drill		
Trip Drill		

STOP Cards

Number of Reports: 36

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	02:00	2.00	32"	D	03	P	Cont' drill 32" hole section f/ 163m to 183m (TD) w/ 1050 gpm / 1650 psi / 15 kips / 50 rpm / 3 - 8 kft-lbf.
02:00	02:30	0.50	32"	D	07	P	Pump 60 bbls GG pill + 60 bbls PHG and circulate out w/ 950 gpm & 1250 psi & 20 rpm. Take survey on bottom, zero angle.
02:30	03:30	1.00	32"	D	06	P	POOH f/ 183m to 115m & RIH back to Btm in good hole condition. Obs no hole fill up.
03:30	04:30	1.00	32"	D	07	P	Pump 100 bbls GG pill & circulate out. Spot 400 bbls of 10 ppg PHG pill on btm.
04:30	06:30	2.00	32"	D	04	P	POOH 32" BHA f/ 183m to Surface. R/B BHA. Observe one cone of hole opener balled up. Two nozzles of hole opener plugged. No bit balled up.
06:30	07:00	0.50	32"	E	18	P	R/U AL-Ghaith 26" CP handling tools.
07:00	12:30	5.50	32"	E	18	P	HPJSM, P/U 26" CP Shoe Jt, Check drainage, Ok, RIH 26" uncoated CP (169.4 ppf, X-56, SWIFTDW2) to 69.1m, P/U & M/U 26" coated CP and continue RIH to 165m. (Joint # 6,7,8,9,10,11,12). *** Perfect hole condition while 26" CP running. *** Totally 5 uncoated + 7 coated 26" CP (169.4 ppf, X-56, SWIFT DW2) *** Install 26" x 32" Bow Centralizers on Joints # 1,2 & 5. *** Install Platform Rigid Centralizers on Joints # 8, 9, 10, 12. *** Fill up every 26" C.P Joint with sea water once at sea level while running.
12:30	13:00	0.50	32"	E	18	P	Change out 26" CP elevator to 5 7/8" DP elevator. P/U Oil State Lynx SA2 Running Tool Assy & M/U to 26" CP.
13:00	14:00	1.00	32"	E	18	N	Connect TDS. Try to make circulation. Observe shoe got plugged. Apply max 950 psi pressure and work on string, no success. Decided to land conductor, then stablish circulation with stinger.
14:00	15:30	1.50	32"	E	18	P	RIH on CP RT to 178m. (26" CP Shoe). M/U adjustable clamp around 26" CP on the jacket.
15:30	16:00	0.50	32"	E	18	P	Slack off weight on landing frame, Observe good centralization of Joint #12 inside 5th level Platform funnel (The 1st platform level above water level), Disconnect Oil State Lynx SA2 Running Tool, POOH & R/B same.
16:00	17:00	1.00	32"	H	04	P	RIH Cementing stinger assembly (CMT stinger + 1 Pup Joint with centralizer on coupling) on 5 7/8" DP. P/U & M/U one jt 5 7/8" Pup Joint + CMT assembly (T-sub, Lo-torque valve & FOSV), RIH to top of shoe.
17:00	18:00	1.00	32"	H	07	N	Circulate high rate on top of shoe. Stop pumping, sting into 26" shoe @ 178m, Slack off 10 Kips. Try to unplug shoe by applying pressure in steps. Shoe got free @ 1950 psi.
18:00	18:30	0.50	32"	H	07	P	Fill up 26" CP x 5 7/8" DP annulus w/ sea water. Circulate w/ 250 gpm. Monitor & check annulus level, Static. Confirm all seals ok.

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Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
18:30	20:30	2.00	32"	H	09	P	L/U to Mehran Cementing unit, Flush same with sea water, Pressure test to 2000 psi / 5min, Ok, Mix and pump 263 bbls of 15.8 ppg cement slurry (Class G) at 5 - 6 BPM / 200 - 400 psi, Displace w/ 12 bbls SW (3 bbls short). Sting out. Float holding, Ok. P/U string 3m, Circulate at 1000 gpm. *** Consume 61 MT of Bulk Cement Class "G" .
20:30	21:00	0.50	32"	H	04	P	POOH CMT stinger assembly on 5 7/8" DP to surface, rack back same. >> Prepare for Skid to well SPD19B-05.

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
183.00	Sea Water	8.60				
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.50	8.	345	8.0	1,014.0	21.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note

Mud Additives

Description	Consumed	Cum Rec	On Loc

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
1	26	Varel	L115J	1216681	Mill Tooth	115	20/20/20/18
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
1.17	6.20	14,880.00	104.50	183.00	78.50	12.7	1-1-NO-G-E-I-NO-TD
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
1,050	1,050	1,650.0	1,650.0	10	10	40	40
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
9.61	6.20	100.00	0.00		165		

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
163.00	183.00	1.20	6.0	1,050	40	1,650.0	10	165

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
XOS	7	2.810	0.79	96.85
5 Jt 8 1/4" DC	8 1/4	2.875	46.99	96.06
XOS	9 5/8	3.000	1.22	49.07
2 Jt 9 1/2" DC	9 1/2	3.000	18.84	47.85
26" S. Stabilizer	26	3.125	2.05	29.01
9 1/2" NMDC	9 1/2	4.000	9.45	26.96
9 1/2" MWD Sub	9 1/2		2.15	17.51
9 1/2" NMDC	9 1/2	4.000	9.46	15.36
9 1/2" Restrictor Sub	9 1/2	4.250	0.92	5.90
Bit Sub (With HPOGC valve inside)	9 1/2		0.98	4.98
32" Hole Opener	32		3.39	4.00
26" Mill Tooth Bit	26		0.61	0.61

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**

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General Remarks

Comment

>>> Requirements: NIL

>>> Sent: NIL

>>> Received:

*** Swissco f/ Kish: 50 MT cement + 180 MT DW

*** Morshed f/ Kish: 125 MT DW + 10 no. rigid centralizer + 50 BB bentonite

*** Navis King f/ Kish: 100 MT DW + 50 MT cement

*** Miami f/ Kish: PDF MR#1 + 60 MT cement + AMOS CMT chemicals

*** C-Mariner f/ Kish: 2 sets of Conductor Pipe

*** Yaghoub f/ Kish: 1 no. COR 32" Hole Opener + 6 no. 16/32" nozzle for HO